

Deep Residual Learning for Image Recognition

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Abstract

Deeper neural networks are more difficult to train. We present a residual learning framework to ease the training of networks that are substantially deeper than those used previously. We explicitly reformulate the layers as learning residual functions with reference to the layer inputs, instead of learning unreferenced functions. We provide comprehensive empirical evidence showing that these residual networks are easier to optimize, and can gain accuracy from considerably increased depth. On the ImageNet dataset we evaluate residual nets with a depth of up to 152 layers -- 8 times deeper than VGG nets but still having lower complexity.

1. Introduction

Deep convolutional neural networks have led to a series of breakthroughs for image classification. Driven by the significance of depth, a question arises: Is learning better networks as easy as stacking more layers? An obstacle to answering this question was the notorious problem of vanishing/exploding gradients, which hamper convergence from the beginning.

When deeper networks are able to start converging, a degradation problem has been exposed: with the network depth increasing, accuracy gets saturated and then degrades rapidly. Such degradation is not caused by overfitting, and adding more layers to a suitably deep model leads to higher training error.

In this paper, we address the degradation problem by introducing a deep residual learning framework. Instead of hoping each few stacked layers directly fit a desired underlying mapping, we explicitly let these layers fit a residual mapping.

Eq: $H(x) = F(x) + x$, where $F(x)$ is the residual mapping to be learned

3. Deep Residual Learning

3.1 Residual Learning. We consider $H(x)$ as an underlying mapping to be fit by a few stacked layers, with x denoting the inputs to the first of these layers. If one hypothesizes that multiple nonlinear layers can asymptotically approximate complex functions, it is equivalent to hypothesize that they can asymptotically approximate residual functions.

3.2 Identity Mapping by Shortcuts. We adopt residual learning to every few stacked layers. A building block is defined as: $y = F(x, \{W_i\}) + x$. The operation $F + x$ is performed by a shortcut connection and element-wise addition. The shortcut connections introduce neither extra parameter nor computation complexity.

4. Experiments

We evaluate on the ImageNet 2012 classification dataset consisting of 1000 classes. We show that: 1) Our extremely deep residual nets are easy to optimize, but the counterpart 'plain' nets exhibit higher training error when the depth increases; 2) Our deep residual nets can easily enjoy accuracy gains from greatly increased depth, producing results substantially better than previous networks. A 152-layer residual

network achieves a top-5 error of 3.57% on the ImageNet test set, winning the 1st place in the ILSVRC 2015 classification task.

5. Conclusion

We have presented a residual learning framework that allows for training of substantially deeper networks than were previously feasible. This work has fundamental significance for many other visual recognition tasks and non-visual problems where deep networks are beneficial.